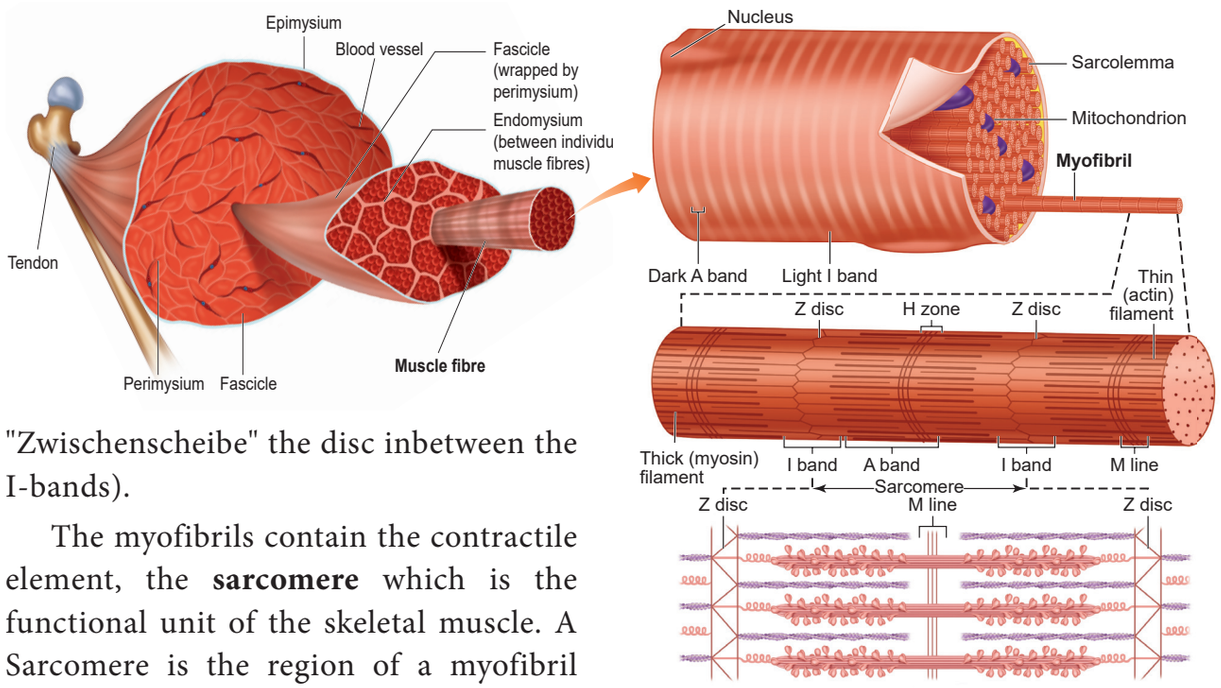


Figure 9.1 Organizational level of a skeletal muscle

"Zwischenscheibe" the disc inbetween the I-bands).

The myofibrils contain the contractile element, the **sarcomere** which is the functional unit of the skeletal muscle. A Sarcomere is the region of a myofibril between two successive Z-discs. It contains an A-band with a half I-band at each end. Inside the sarcomere two types of filaments are present namely the **thick** and **thin filaments**.

The thick filaments extend the entire length of the A-band, the thin filaments extend across the I-band and partly into the A-band. The invagination of the sarcolemma forms transverse tubules **(T–tubules)** and they penetrate into the junction between the A and I-bands.

Muscle Terminology

General Term	Muscle Equivalent
Cell	Muscle fibre/ Myofibril
Plasma membrane	Sarcolemma
Cytoplasm	Sarcoplasm
Endoplasmic reticulum	Sarcoplasmic reticulum

9.4 Structure of contractile proteins

Contraction of the muscle depends on the presence of contractile proteins (Figure 9.2) such as **actin** and **myosin** in the myofilaments. The thick filaments are composed of the protein myosin which are bundled together whose heads produce at opposite ends of the filament. Each myosin molecule is made up of a monomer called meromyosin. The meromyosin has two regions, a globular head with a short arm and a tail. The short arm constitutes the heavy meromyosin (HMM). The tail portion forms the light meromyosin (LMM). The head bears an actin-binding site and an ATP- binding site. It also contains ATPase enzyme that split ATP to generate energy for the contraction of muscle. The thin filaments are composed of two interwined actin molecules. Actin has polypeptide subunits called globular actin or G–actin and filamentous form or F-actin. Each thin filament is made of two F-actins helically wound to each other. Each F-actin is a polymer of monomeric G-actins. It also contains a binding site for myosin. The thin filaments also

TN_GOVT_XI_Bio Zoology_Ch09.indd 153 09-01-2020 5.30.51 PM